Multiloss Adversarial Attacks for Multimodal Remote Sensing Image Classification

Qi Hu, Zhidong Shen, Zongyao Sha, and Weijie Tan

***Abstract*— The challenge of classifying multimodal remote sensing images has garnered significant interest in light of the growing diversity of remote sensing image data modalities. Current studies primarily concentrate on increasing the classification task's accuracy by improving the fusion strategy or incorporating auxiliary architectures. However, there is currently a lack of research in the area of adversarial attacks for multimodal remote sensing image classification models as compared to the study of unimodal classification models. To overcome this issue, our research first investigates the adversarial robustness of multimodal remote sensing image classification under different fusion strategies, which adopts the improved classical adversarial attack methods to test the adversarial robustness of multimodal remote sensing image classification model architectures with three different fusion strategies. Then, a new multimodal adversarial attack method is proposed for the multimodal model, which adopts balanced perturbation loss and cooperative adversarial loss, in which the balanced perturbation loss is used to balance the level of perturbation of different modalities, and the cooperative adversarial loss is used to reduce the conflict of different modality perturbations on the classification result. By combining balanced perturbation loss and cooperative adversarial loss to attack multimodal models, the cooperation between modalities is continuously optimized. Finally, the study demonstrates the weak adversarial robustness of the multimodal remote sensing image classification model, the robustness of which is easily influenced by the fusion strategies and the attack methods. Additionally, a better attack effect is obtained by the multimodal multiloss cooperative adversarial attack method proposed in this article.**

***Index Terms*— Adversarial attack, multimodal, remote sensing image classification, robustness.**

I. Introduction

REMOTE sensing image classification is a pixel-level classification task, which assigns a corresponding category label to each pixel of a remote sensing image according to each pixel's different spectral brightness values and the spatial structure relationship between pixels. This task is one of the most essential tasks in the research of remote sensing technology because it can recognize the attributes of ground objects and provide relevant geographic information for land-cover mapping, emergency response, forest detection, and other applications [1], [2], [3]. With the development of deep learning, this technique is likewise widely used in remote sensing image classification due to its powerful discriminative feature representation [4]. However, while deep learning achieves excellent results in this task, it also suffers from weaknesses that are susceptible to adversarial samples. It is feasible to make the classification model mispredict the classified pixels by adding carefully planned tiny perturbations to the remote sensing image input, which are undetectable by the human eye.

Some works [5], [6], [7], [8] have investigated the adversarial robustness against adversarial attack methods for remote sensing images with different modalities of inputs with different network structures and different classification tasks, showing that the model is vulnerable to adversarial samples in any combination of cases. With the deepening of the research on remote sensing adversarial attacks, some adversarial defense methods have also been proposed, like input filtering [9], adversarial training [10], and other methods have been applied to the field of remote sensing image classification to achieve a certain defense effect and improve the model robustness.

To enhance the accuracy and robustness of the remote sensing image classification task, multimodal fusion architectures have been proposed for this task. Compared with unimodal inputs, the fusion of information from multimodal data can provide redundant and complementary information to each other to create more effective and robust classification models. Several studies have focused on different modal inputs and intermodal fusion strategies. There are many types of remote sensing image data, such as hyperspectral images (HSIs), multispectral images (MSs), synthetic aperture radar (SAR), light detection domain ranging (LiDAR), and so on. HSIs and MSs can provide rich spatial information about the scene in high resolution, while LiDAR and SAR images contain more height information about the scene. Some popular combinations of multimodal remote sensing image dataset types are HSI-MSI [11], HSI-LiDAR [12], and HSI-SAR [13]. Depending on the level of fusion, the

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intermodal fusion strategies involve early fusion [14], [15], [16], [17], [18], middle fusion [19], [20], [21], [22], and late fusion [23], [24], [25]. Early fusion is pixel-level fusion, that is, the direct fusion of different modal data at the input level; middle fusion is feature-level fusion, in which the extracted features of multiple modalities are fused by a fusion network to form a unified representation, and then predicted for classification; and late fusion is the decision-level fusion, which fuses the classifications predicted by modality-independent classification networks and takes them as the final classification results.

However, although multimodal remote sensing image classification has been widely researched, most of them only focus on how to design better fusion strategies to achieve higher accuracy of the models and often neglect the adversarial robustness of multimodal remote sensing image classification models. Few studies have systematically analyzed the adversarial robustness performance of multimodal remote sensing image classification models and designed specialized adversarial attack and defense schemes.

To analyze the robustness of multimodal remote sensing image classification models, this article investigates the multimodal classification models with three fusion strategies using different adversarial methods and attack strategies. The attack strategies include unimodal attack and multimodal attack, which are the adversarial attack against single modal perturbation and the adversarial attack against multiple modal perturbations, respectively. The robustness of multimodal classification models with different fusion strategies against adversarial attacks is investigated through experiments. Meanwhile, this article proposes an adversarial attack method for attacking multimodal remote sensing image classification models, which adopts an optimization-based strategy and considers the following aspects when generating multimodal perturbations: 1) the image after adding the perturbation should be close to the original image; 2) the perturbation strengths between multiple modalities should be similar, preventing a single modal from playing a dominant role and fully mobilizing the multisource modal attack; and 3) the perturbations between modalities should produce similar perturbation distributions in the classification results, preventing different modal perturbation attacks from conflicting with each other to counteract each other; the perturbations optimized by this method can maintain a high success rate of attack under the smaller perturbations.

The study's significant contributions are as follows.

1) The performance of multimodal remote sensing image classification models with different fusion strategies is analyzed under unimodal attack and multimodal attack of different adversarial methods, which helps to understand the adversarial robustness of the models.
2) An optimization-based adversarial attack method is proposed, which constrains the similarity of the perturbation strengths of different modalities and the similarity of the impacts on the original classification result after the modalities' respective perturbations, and mobilizes the cooperative attack among modalities.
3) Experiments are conducted on different multimodal fusion architectures and two popular datasets, which show that the optimization method proposed in this article achieves stronger attack performance and can obtain more effective attack results under low perturbation.

II. Related Works

A. Remote Sensing Image Classification

There are many shallow models for multimodal remote sensing image classification such as morphological operators and subspace learning. Since deep-learning technology has a strong ability to generate discriminative features, in recent years, many researchers have adopted deep-learning technology to extract and classify features from multimodal remote sensing images. The fusion strategy between modalities is the key to deep-learning-based multimodal fusion networks. Different fusion strategies can be categorized into early, middle, and late fusion, according to the fusion levels. Early fusion fuses multiple independent modal data into a single data and inputs it into the classification model. Hong et al. [14] proposed fusing different modalities at the pixel level at the input and demonstrated that early fusion may inhibit the interaction of information within and between modalities, increasing the risk of overfitting. In [15], it directly treated the LiDAR data as an additional spectral band of HSI data. Then, it fed the spliced fused data into the network for feature learning and classification. Kulkarni and Rege [16] covered different pixel-level methods for fusing HSI and SAR images. Li et al. [17] analyzed the performance and challenges of pixel-level image fusion methods like multiscale decomposition and sparse representation in four major areas: remote sensing, medical diagnostics, surveillance, and photography. Liu et al. [18] introduced more than a dozen pixel-level image fusion methods based on convolutional neural networks (CNNs), convolutional sparse representation, and stacked autoencoder using deep-learning techniques and formulated a generalized framework and evaluation metrics. Middle fusion transforms different modal data into features before fusing them in the intermediate layer of the model, which is flexible in terms of where and how to fuse, thus is the method used in most studies. Chen et al. [26] first used two two-stream CNNs to extract feature representations of the HSI and the LiDAR data, respectively. Then, they used the fusion network to further fuse the extracted features to obtain the joint representation and further classify them. Its architecture is shown in Fig. 1. Xu et al. [10] analyzed the effect of the fusion of HSI and high-resolution visual images using the same CNN dual-branch architecture. Zhu et al. [21] proposed a similar dual-center network SDCN, which first trained the HSI network and then added similar network branches to the trained network for extending the LiDAR information to the HSI model. In [27], it proposed multimodal deep learning with cross-weighting (MDL-CW) which was to interactively represent multimodal features to obtain more effective information fusion. Late fusion is achieved by inputting different modal data into different classifiers to obtain the classification results and then fusing these classification results. Hong et al. [19] introduced

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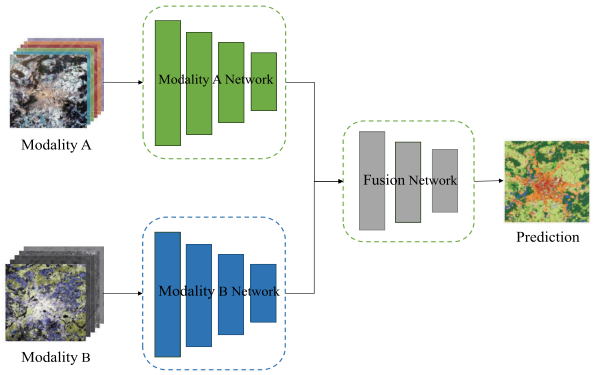


Fig. 1. Multimodal remote sensing image classification architecture for middle fusion.

decision-level fusion, which fully utilizes the integration characteristics of different deep neural networks. In [24], four SVM classifiers were trained using spectral features, spatial features, elevation features, and graph fusion features as inputs, respectively, and the four classification maps were fused by weighted majority voting to obtain the final classification map. Zhao et al. [25] used a decision-level fusion strategy to fuse the classification results of linear and nonlinear features in an HSI classification task.

B. Adversarial Attack

The nonlinear characteristics and overfitting of deep-learning models lead to the vulnerability of the model to specific tiny perturbations in the picture. Thus, adversarial attack is a major threat to deep-learning models. Szegedy et al. [28] presented a pioneering work that first found that deep-learning networks are vulnerable to small perturbations and proposed to use box-constrained L-BFGS to optimize adversarial perturbations to spoof the network. Based on this, many studies have proposed various white-box (network structure and parameters are known) attack methods to effectively attack image classifiers, including the gradient-based fast gradient sign method (FGSM) [29], the iterative FGSM [30], the projection gradient descent (PGD) [31], Jacobi-based saliency map attack (JSMA) [32], decision surface-based DeepFool [33], Universal Adversarial Perturbation Attack UAP [34], optimization-based Carlini and Wagner (C&W) attack [35], multivariate input iterative attack [36], momentum-based iterative method (MIM) [37], generative adversarial networks (GANs)-based attack AdvGAN [38], and so on.

Some corresponding adversarial attack methods and defense methods have also been proposed in the field of remote sensing. Cheng et al. [39] applied the adversarial attack to the remote sensing image target tracking task by introducing adversarial perturbations generated by attentional feature loss, attentional confidence loss, background dispersal loss, and total variation loss constraints to spoof the target tracking model of UAV remote sensing images. To enhance the physical realizability of the adversarial samples, [40] proposed an advanced patch attack for optical remote sensing image target detection. By selecting more aggressive patch subblocks through first-order differencing and causing the target detection frame to move away through bounding box drift loss, the method proposed a more threatening patch effect. Li et al. [41] presented a generic white-box attack method with local perturbation and adaptive global attack, which realized the generation of generic perturbations on adversarial detectors with different structures to deceive the models. Shi et al. [42] presented a multifeature collaborative adversarial network for generating multimodal adversarial samples based on the adversarial generative network, which performed better than the classical adversarial attack methods on multimodal models. Some adversarial defense methods have also been proposed along with the research on adversarial attack methods: [39] designed a new training framework to generate adversarial samples with random types of attacks added by perturbation seeking GAN and trained the classifier with clean samples and various types of adversarial samples to improve the robustness of the classifier. Tu et al. [43] improved the robustness of the model by expanding the convolutional learning and context-aware learning modules to construct detailed global context information to prevent adversarial attacks.

Unlike individual visual adversarial attacks, we study multimodal attacks under multimodal remote sensing image classification, which generates adversarial samples for multiple modal inputs to jointly perturb the multimodal classification model.

III. Adversarial Robustness for Multimodal Remote Sensing Image Classification

A. Attack Analysis and Implementation

1) Remote Sensing Image Classification: The multimodal remote sensing image classification task performs feature extraction and classification by fusing two or more types of input data. In this article, we focus on multimodal fusion based on HSI and LiDAR modalities. According to the different network stages in which the two modalities are fused, the fusion strategies can be classified into three categories: early, middle, and late fusion, as shown in Fig. 2.

In Fig. 2, the feature transformation function of the lth layer of the fusion network is defined as $G_l(\cdot)$, and the HSI and LiDAR modalities are defined as M_H and M_L, respectively. The network features of the two branches at the lth layer are $f_l^{M_H}$ and $f_l^{M_L}$, respectively. The fusion operations, such as weighted sum, averaging, and splicing, are defined as $\oplus$ so that the network models of the different fusion levels can be described as follows.

1) *Early Fusion:* Early fusion fuses raw or simply processed HSI and LiDAR data together, assuming a certain L-layer neural network based on the early fusion framework, the early fusion network shown in Fig. 2(a) can be defined by the following equation:

$$f_L = G_L\left(G_{L-1}\left(\cdots G_l\left(\cdots G_2\left(G_1\left(f_0^{M_H} \oplus f_0^{M_L}\right)\right)\right)\right)\right). \tag{1}$$

In the early fusion stage, the network learns the joint features of several modalities, fully utilizing the information from the original data without losing any original information. However, the early fusion model

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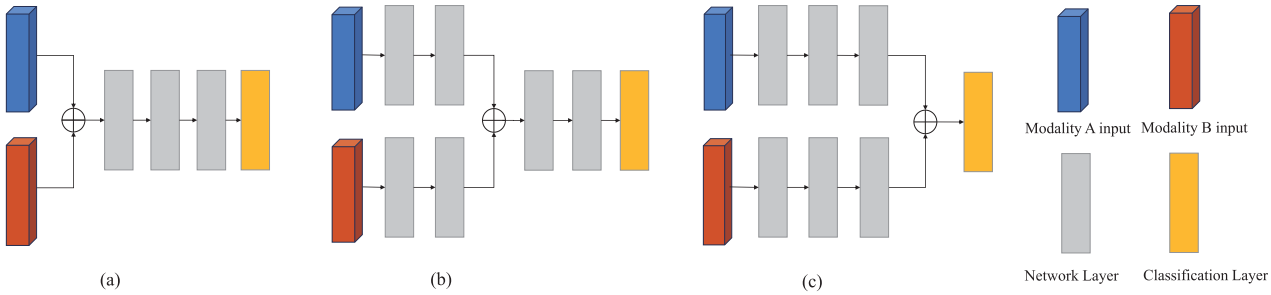


Fig. 2. Architecture of different fusion strategies. (a) Early fusion. (b) Middle fusion. (c) Late fusion.

does not have the flexibility to migrate to different types of data, for example, when the inputs are replaced with new modality data or the dimensionality of the input features is increased or decreased, the early fusion network needs to be completely redesigned and retrained.

2) *Middle Fusion:* Middle fusion utilizes a neural network to extract contextual features from each modality data. Then, it combines the feature representations of different modalities in a middle hidden layer, fusing the features from each modality to generate new features for further processing.

The layer that fuses intermediate features is defined as the middle fusion layer $G_{l^*}(\cdot)$. The middle fusion can only be performed once in this layer. The middle fusion network shown in Fig. 2(b) can be defined by the following equation:

$$f_L = G_L\Big(\cdots G_{l^*+1}\Big(G_{l^*}^{M_H}\Big(\cdots G_1^{M_H}\Big(f_0^{M_H}\Big)\Big) \oplus G_{l^\star}^{M_L}\Big(\cdots G_1^{M_L}\Big(f_0^{M_L}\Big)\Big)\Big)\Big). \quad (2)$$

Middle fusion methods are very flexible, but given a specific network architecture, it can be challenging to find a suitable way of fusion, often relying on a large amount of training data to find the most important subset of features. Compared to early fusion, middle fusion methods can generate a smaller information space with less computational load.

3) *Late Fusion:* Late fusion combines the results of each modality network into a global decision output, as shown in Fig. 2(c). The late fusion network can be represented by the following equation:

$$f_L = G_L^{M_H}\Big(G_{L-1}^{M_H}\Big(\cdots G_1^{M_H}\Big(f_0^{M_H}\Big)\Big)\Big) \oplus G_L^{M_L}\Big(G_{L-1}^{M_L}\Big(\cdots G_1^{M_L}\Big(f_0^{M_L}\Big)\Big)\Big). \quad (3)$$

Late fusion is highly flexible and modular. Only the domain-specific network needs to be trained when adding a new modality without affecting the other networks. At the same time, it has lower complexity and less computational load and communication resources. In addition, late fusion can standardize the interfaces of fusion methods and does not require in-depth knowledge of the deep learning feature extraction methods involved. However, it also discards the rich intermediate features across modalities, which are often very useful in fusion.

2) Multimodal Adversarial Attack: In this article, the adversarial attacks against multimodal remote sensing classification tasks are categorized into unimodal attacks and multimodal attacks according to the number of perturbed modalities. Since the model discussed in this article has two modality inputs, HSIs and LiDAR, a unimodal attack against this model is to attack the model by perturbing HSI or LiDAR data alone, while a multimodal attack is to attack the model by perturbing HSI and LiDAR modality inputs at the same time.

a) Unimodal attack: We mainly use the following classical adversarial attacks for the unimodal attack on multimodal models.

1) *FGSM:* This method is a fast gradient descent method based on gradient proposed by Goodfellow et al. [29]. By adding perturbation to the image in the direction of the model's gradient, the model's loss function value is changed so that the model produces an erroneous wrong classification output. Its formula for generating adversarial samples is as follows.

$$x' = x + \varepsilon \mathrm{sgn}(\nabla_x L(x, y)) \quad (4)$$

where x is the origin image, x' is the adversarial sample after adding the perturbation, and y is real label. $\nabla_x L(x, y)$ describes the gradient of the loss function $L(x, y)$ at point x, which is processed by the sign function $\mathrm{sgn}(\cdot)$ to become the gradient direction. The optimized objective function is as follows:

$$\max L(x, y), \quad \text{s.t. } \left\|x' - x\right\|_\infty \le \varepsilon. \quad (5)$$

Equation (5) searches for adversarial samples x' unclassified as y in the neighboring balls ε of the origin image x, and the adversarial samples can trick the deep-learning model with slight perturbations. The gradient only needs to be computed once in FGSM, and it stacks data more quickly.

2) *Projected Gradient Descent (PGD):* PGD [31] is actually an iterative method of FGSM. The quality of the adversarial sample is significantly improved by introducing multiple iterations. The relative formula is as follows:

$$x_{i+1} = \mathrm{clip}_{x,\varepsilon}\big(x_i + \mu \mathrm{sgn}\big(\nabla_{x_i} L(x_i, y)\big)\big) \quad (6)$$

where $x_0 = x$ and μ is the iterative step size. $\mathrm{clip}(\cdot)$ is used to restrict the constraints x_i to the neighborhood ε of the legal image x.

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3) *Momentum-Based Iterative Method (MIM):* MIM [37] incorporates a momentum term in the iterative process to further stabilize the update direction and reduce the occurrence of local minima. The relative formula is as follows:

$$\begin{cases} g_{t+1} = \mu \cdot g_t + \dfrac{\nabla_{x_t^*} J\left(x_t^*, y\right)}{\| \nabla_{x_t^*} J\left(x_t^*, y\right)\|_1} \\ x_{t+1}^* = x_t^* + \alpha \mathrm{sgn}(g_{t+1}) \end{cases} \tag{7}$$

where g_t is the momentum term, μ is the attenuation factor that moderates correlation, and α is the iterative step size.

4) *C&W:* C&W [35] is an optimization-based attack method proposed by C&W based on different adversarial perturbation metric paradigms and adversarial loss metric functions, which is formulated as follows:

$$\min \left\|x' - x\right\|_2^2 + cf\left(x', y\right) \tag{8}$$

where the first term of the optimization function is the perturbation loss. The second term $f(\cdot)$ is the adversarial loss and c is the hyperparameter that balances the perturbation loss and the adversarial loss.

b) Multimodal attack: This attack perturbs two modalities of the multimodal model simultaneously, and we improve four unimodal adversarial attack methods of FGSM, PGD, MIM, and C&W into multimodal methods, and the specific implementation details are shown in (9)–(13).

1) *Multimodal FGSM Attack:* This method simultaneously perturbs the inputs at both HSI and LiDAR modalities, and the perturbations are generated by a one-step inverse gradient perturbation, respectively,

$$\begin{cases} x'_H = x_H + \varepsilon \mathrm{sgn}\left(\nabla_{x_H} L(x_H, x_L, y)\right) \\ x'_L = x_L + \varepsilon \mathrm{sgn}\left(\nabla_{x_L} L(x_H, x_L, y)\right). \end{cases} \tag{9}$$

2) *Multimodal PGD Attack:* This method performs an iterative gradient attack in both HSI and LiDAR modalities

$$\begin{cases} x_{H,i+1} = \mathrm{clip}_{x_H,\varepsilon}\left(x_{H,i} + \mu \mathrm{sgn}\left(\nabla_{x_{H,i}} L\left(x_{H,i}, x_{L,i}, y\right)\right)\right) \\ x_{L,i+1} = \mathrm{clip}_{x_{L,\varepsilon}}\left(x_{L,i} + \mu \mathrm{sgn}\left(\nabla_{x_{L,i}} L\left(x_{H,i}, x_{L,i}, y\right)\right)\right). \end{cases} \tag{10}$$

3) *Multimodal MIM Attack:* This method introduces momentum terms into the gradient updating process and employs an iterative gradient-based attack to perturb the HSI and LiDAR modalities separately

$$\begin{cases} g_{H,t+1} = \mu \cdot g_{H,t} + \dfrac{\nabla_{x_{H,i}} L\left(x_{H,i}, x_{L,i}, y\right)_x}{\left\|\nabla_{x_{H,i}} L\left(x_{H,i}, x_{L,i}, t\right)\right\|_1} \\ g_{L,t+1} = \mu \cdot g_{L,t} + \dfrac{\nabla_{x_{H,i}} L\left(x_{H,i}, x_{L,i}, y\right)_x}{\left\|\nabla_{x_{H,i}} L\left(x_{H,i}, x_{L,i}, t\right)\right\|_1} \end{cases} \tag{11}$$

$$\begin{cases} x_{H,i+1} = \mathrm{clip}_{x_H,\varepsilon}\left(x_{H,i} + \mu \mathrm{sgn}\left(g_{H,t+1}\right)\right) \\ x_{L,i+1} = \mathrm{clip}_{x_L,\varepsilon}\left(x_{L,i} + \mu \mathrm{sgn}\left(g_{L,t+1}\right)\right). \end{cases} \tag{12}$$

4) *Multimodal C&W Attack:* This method is based on optimizing the solution of the perturbations while considering the size of the perturbations added to the two modalities and the adversarial loss of the multimodal network

$$\min \left\|x'_H - x_H\right\|_2^2 + \left\|x'_L - x_L\right\|_2^2 + cf(x_H, x_L, y). \tag{13}$$

B. Experiments

1) Datasets and Indicators: We test the adversarial robustness of the multimodal model on two HSI and LiDAR fusion datasets.

a) Houston2013 dataset: It was obtained using the ITRES CASI1500 image sensor on the campus of the University of Houston and its adjacent rural areas in Houston, Texas, USA, and was released by IEEE GRSS [44]. The dataset is made up of one band of 349 × 1905 pixel LiDAR data and 144 bands of HSI images ranging from 364 to 1064 nm. There are 15 different classes in this dataset.

b) Trento dataset: The Trento dataset was taken in a rural area of Trento, Italy. The HSI data consists of 63 spectral bands converging the wavelength of 0.42 to 0.99 μm, and the LiDAR data consists of one band. There are six different classes.

The pixel-level remote sensing image classification performance evaluation metrics usually contain three metrics including overall accuracy (OA), average accuracy (AA), and κ coefficient which are used to quantify the classification performance from different perspectives, and they can be expressed by the following equations:

$$\mathrm{OA} = \frac{N_c}{N_a} \tag{14}$$

$$\mathrm{AA} = \frac{1}{C} \sum_{i=1}^{C} \frac{N_c^i}{N_a^i} \tag{15}$$

$$\kappa = \frac{\mathrm{OA} - P_e}{1 - P_e} \tag{16}$$

where N_c denotes the number of correctly classified samples and N_a denotes the total number of samples, respectively. OA defines the proportion of correctly categorized pixels to the overall volume of pixels in the test set. N_c^i and N_a^i correspond to N_c and N_a for each category, respectively. AA refers to the accuracy average in all categories. The κ represents the hypothesized probability of chance consistency. The formula of P_e is as follows:

$$P_e = \frac{N_r^1 \times N_p^1 + \cdots N_r^i \times N_p^i + \cdots + N_r^C \times N_p^C}{N_a \times N_a} \tag{17}$$

where N_r^i and N_p^i stand for the corresponding numbers of genuine samples and predicted samples in each category, respectively. κ, which ranges in value from -1 to 1, is a measurement of the consistency between the model's predictions and the actual classification outcomes.

2) Experimental Settings: To analyze the robustness of multimodal models in a more refined way, we fully compare several models. We first investigate the accuracy and adversarial robustness of the unimodal models HSI network and LiDAR network, which are used as two baselines to

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TABLE I

ACCURACY OF HSI AND LIDAR UNIMODAL MODELS UNDER THE HOUSTON DATASET AND THE TRENTO DATASET USING DIFFERENT ATTACK METHODS. ✔ REPRESENTS THE MODALITY IS CLEAN, ✘ REPRESENTS THE MODALITY IS UNDER ATTACK, ✔ HSI REPRESENTS A CLEAN HSI UNIMODAL MODEL, AND ✘ HSI REPRESENTS AN ATTACKED HSI UNIMODAL MODEL

Dataset	Attack	✔ HSI	✔ LIDAR	✘HSI	✘LIDAR
Houston	FGSM	0.8723	0.4972	0.2366	0.3163
	PGD			0.0871	0.2965
	MIM			0.0691	0.2893
	C&W			0.0584	0.2578
Trento	FGSM	0.9705	0.8771	0.3563	0.4653
	PGD			0.1615	0.3668
	MIM			0.1385	0.3294
	C&W			0.0732	0.2812

explore the effect of multimodal fusion on model accuracy and robustness. Meanwhile, we investigate three fusion strategies on the multimodal fusion model, which are early fusion, middle fusion, and late fusion, and the adversarial attack strategies are categorized into unimodal attack and multimodal attack according to the number of modalities attacked to discuss the robustness of the multimodal model under different fusion strategies and different attack strategies.

All deep-learning models and methods are implemented in the PyTorch framework. In the training process, the ratio of the training set to the validation set is 0.8, a local 7×7 size patch of each pixel is used as the input of the model, and the class probability of that pixel is used as the model output for supervised training of the model. To optimize them, we utilize the Adam algorithm [45] to optimize them. The parameters for the batch size, learning rate, and training steps are, respectively, set to 64, 0.001, and 200.

To fairly compare different attack methods, we uniformly limit the perturbation size ε to 0.003 on the Houston dataset and 0.01 on the Trento dataset. For multimodal PGD and MIM attacks, μ is set to 1/8 of the maximum perturbation limit and the number of iteration rounds is set to 10. For C&W attacks, the hyperparameter c is initialized to 0.01 on the Houston dataset, and 0.1 on the Trento dataset, and is continuously optimized during attack method optimization. The learning rate and number of iterations are set to 0.01 and 50.

3) Results Analysis: We investigate the adversarial robustness of multimodal remote sensing image classification models under different fusion strategies with different adversarial methods and different attack strategies. The different fusion strategies include early fusion, middle fusion, and late fusion. The different attack strategies include unimodal attack and multimodal attack. To better investigate and explain the multimodal robustness, we also investigate the baseline results of two unimodal models, HSIs and LiDAR (i.e., using only the HSI modality and LiDAR modality, respectively), subject to adversarial attacks.

From Table I, it is observed that the classification accuracy of the unimodal HSI model is higher than the unimodal LiDAR model under clean data input. The FGSM, PGD, MIM, and C&W methods in the table significantly reduce the classification effectiveness, where C&W is the most effective attack method among several attacks. Meanwhile, different attack methods attack the two unimodal models separately under the same intensity, and the accuracy of the HSI model is lower than that of the LiDAR model, which indicates that the attack against the HSI data is more effective than the attack against the LiDAR data.

TABLE II

ACCURACY OF EARLY FUSION MULTIMODAL REMOTE SENSING CLASSIFICATION MODELS UNDER THE HOUSTON DATASET AND THE TRENTO DATASET USING DIFFERENT ATTACK METHODS. ACCORDING TO THE NUMBER OF MODALITIES, ATTACKS ARE CATEGORIZED INTO UNIMODAL ATTACKS AND MULTIMODAL ATTACKS. HSI IS A UNIMODAL HSI ATTACK, LIDAR IS A UNIMODAL LIDAR ATTACK, AND HSI-LIDAR IS A MULTIMODAL ATTACK

Dataset	Attack	non	HSI	LIDAR	HSI-LIDAR
Houston	FGSM	0.9044	0.4785	0.9013	0.4760
	PGD		0.2777	0.9013	0.2750
	MIM		0.2476	0.9008	0.2438
	C&W		0.1836	0.9003	0.1797
Trento	FGSM	0.9785	0.6331	0.9780	0.6264
	PGD		0.4643	0.9780	0.4531
	MIM		0.4426	0.9777	0.4284
	C&W		0.2394	0.9776	0.2294

TABLE III

ACCURACY OF MIDDLE FUSION MULTIMODAL REMOTE SENSING CLASSIFICATION MODELS UNDER THE HOUSTON DATASET AND THE TRENTO DATASET USING DIFFERENT ATTACK METHODS

Dataset	Attack	non	HSI	LIDAR	HSI-LIDAR
Houston	FGSM	0.9073	0.5683	0.8957	0.5475
	PGD		0.3751	0.8927	0.3412
	MIM		0.3331	0.8911	0.3056
	C&W		0.2565	0.8853	0.2039
Trento	FGSM	0.9781	0.8579	0.9551	0.7358
	PGD		0.7126	0.9500	0.5505
	MIM		0.6955	0.9419	0.5367
	C&W		0.5610	0.8269	0.3537

TABLE IV

ACCURACY OF LATE FUSION MULTIMODAL REMOTE SENSING CLASSIFICATION MODELS UNDER THE HOUSTON DATASET AND THE TRENTO DATASET USING DIFFERENT ATTACK METHODS

Dataset	Attack	non	HSI	LIDAR	HSI-LIDAR
Houston	FGSM	0.9310	0.5385	0.9237	0.5161
	PGD		0.3260	0.9214	0.2961
	MIM		0.2913	0.9205	0.2593
	C&W		0.2019	0.9218	0.1941
Trento	FGSM	0.9788	0.9001	0.9462	0.7764
	PGD		0.8196	0.9400	0.5929
	MIM		0.7976	0.9329	0.5603
	C&W		0.7265	0.8771	0.2858

We investigate the robustness of different multimodal fusion architectures, that is, early fusion, middle fusion, and late fusion, using unimodal and multimodal attacks, respectively. In terms of clean data accuracy, the multimodal model of each fusion strategy improves the accuracy of the unimodal model. Next, we analyze the results based on different attacks. The results are shown in Tables II–V.

a) Unimodal attack: In models with the same fusion strategy, the adversarial attack against HSIs is more effective in attacking the model accuracy than the adversarial attack against LiDAR. The reasons for this one are that the HSI modality has more bands and the other is because the spectral

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TABLE V
VISUAL QUALITY ASSESSMENT OF MULTIMODAL REMOTE SENSING CLASSIFICATION MODELS UNDER THE HOUSTON DATASET USING DIFFERENT ATTACK METHODS AND DIFFERENT FUSION STRATEGIES

Dataset	Model	Attack	HSI			LIDAR			HSI-LIDAR					
			L_∞	L_2	SSIM	L_∞	L_2	SSIM	L_∞-H	L_2-H	SSIM-H	L_∞-L	L_2-L	SSIM-L
Houston	Early	FGSM	0.0030	0.2519	0.9986	0.0030	0.0206	0.9987	0.0030	0.2519	0.9986	0.0030	0.0206	0.9987
		PGD	0.0030	0.2039	0.9991	0.0030	0.0203	0.9988	0.0030	0.2038	0.9991	0.0030	0.0172	0.9992
		MIM	0.0030	0.2402	0.9987	0.0030	0.0230	0.9985	0.0030	0.2401	0.9987	0.0030	0.0200	0.9991
		C&W	0.0030	0.1612	0.9993	0.0030	0.0158	0.9995	0.0030	0.1607	0.9993	0.0022	0.0122	0.9998
	Middle	FGSM	0.0030	0.2519	0.9986	0.0030	0.0207	0.9986	0.0030	0.2519	0.9986	0.0030	0.0207	0.9986
		PGD	0.0030	0.2071	0.9990	0.0030	0.0186	0.9987	0.0030	0.2069	0.9990	0.0030	0.0179	0.9987
		MIM	0.0030	0.2426	0.9987	0.0030	0.0218	0.9983	0.0030	0.2425	0.9987	0.0030	0.0205	0.9985
		C&W	0.0030	0.1614	0.9992	0.0030	0.0132	0.9996	0.0030	0.1634	0.9992	0.0021	0.0119	0.9998
	Late	FGSM	0.0030	0.2519	0.9986	0.0030	0.0207	0.9983	0.0030	0.2519	0.9986	0.0030	0.0207	0.9983
		PGD	0.0030	0.2049	0.9991	0.0030	0.0188	0.9986	0.0030	0.2049	0.9991	0.0030	0.0186	0.9987
		MIM	0.0030	0.2394	0.9988	0.0030	0.0220	0.9979	0.0030	0.2394	0.9988	0.0030	0.0212	0.9982
		C&W	0.0030	0.1651	0.9993	0.0030	0.0138	0.9997	0.0030	0.1597	0.9993	0.0020	0.0112	0.9998

data is more conducive to image classification. We also find that the addition of another modality in the multimodal model enhances the robustness of the unimodal model, but its increased robustness strength is not as strong as it should be with the addition of the clean modality. For example, the accuracy of the middle fusion model with HSI unimodal attack using the FGSM method is 0.5683, the accuracy of the HSI unimodal model subject to HSI unimodal attack using the FGSM is 0.2366, and the accuracy of the clean LiDAR unimodal model is 0.4972. Since the attacked middle fusion multimodal model has a clean LiDAR modality, we can regard the middle fusion model under HSI unimodal attack as the HSI unimodal model under HSI unimodal attack with the addition of an auxiliary architecture of LiDAR clean modality. The addition of the LiDAR clean modality makes the model accuracy go up from 0.2366 to 0.5683, but the increase is not up to 0.4972.

b) Multimodal attack: In the multimodal attacks against models with different fusion architectures, we find that middle fusion is the most robust, followed by late fusion, and the robustness of early fusion is the worst. When both modalities of a multimodal remote sensing classification model are added with adversarial perturbation, its accuracy by the joint multimodal attack is lower than that by the unimodal attack of any modality. This indicates that the strength of the joint multimodal attack is stronger than the unimodal attack, but we find that the accuracy of the joint multimodal attack is very close to that of the HSI unimodal attack, so we consider how to coordinate the adversarial attacks of the two modalities to make the effect of the joint multimodal attack more obvious.

In general, from Tables II–V, we can see that the C&W attack method is the strongest among all the attack methods. In both unimodal attack and multimodal attack scenarios, the accuracy of the model decreases the most, and the added perturbation quality evaluation values such as L_∞ and L_2 are the smallest, and the structural similarity index measure (SSIM) is the closest to 1, that is, the image after the perturbation is more similar to the original image. The comprehensive attack effect is C&W, PGD, MIM, and FGSM in order of strength to weakness.

c) Comprehensive analysis: We investigate the influence of different perturbation sizes ε on the effects of unimodal and multimodal attacks. As shown in Fig. 3, we study the influence of different perturbation sizes on HSI, LiDAR unimodal attacks, and HSI-LiDAR multimodal attacks in the early fusion model of the Houston dataset. It can be seen that with the increase of the perturbation size, the classification accuracy of the attacked models will continue to decrease, and with the decrease of the model accuracy, the impact of increasing the perturbation size on the attack effect will be smaller, so the declining curve will be more gentle. At the same time, the figure also verifies that the effect of the multimodal attack mentioned above is stronger than that of the unimodal attack, and the HSI modality attack is stronger than the LiDAR modality attack. The intensity of the method attack increases from FGSM, PGD, MIM, and C&W.

To further study the difference between the effects of unimodal attacks and multimodal attacks, we analyze the results of models under various attacks, as shown in Fig. 4. Fig. 4 shows the result maps of HSI and LiDAR unimodal attacks and HSI-LiDAR multimodal attacks on the Houston and Trento datasets of the late fusion model, respectively. In the Houston dataset, when the PGD method is used for unimodal attacks, HSI modality is more dominant in road and parking lot1 categories and LiDAR modality is more dominant in healthy grass, synthetic grass, and running track. The effect of combining the two unimodal attacks makes the attack effect more significant. Similarly, on the Trento dataset, unimodal attacks using the PGD method are more dominant in attacking the HSI modality apple trees category, and LiDAR modality buildings, wood, and vineyard. Multimodal attacks combine the advantages of each single modality. You can attack completely successfully in almost every category.

IV. MULTIMODAL MULTILOSS COOPERATIVE ATTACK METHOD

A. Analysis of Multimodal Attack Bottleneck

1) Conflict Between the Attack Effects of Different Modalities: To attack a multimodal model, the attack effect of perturbing multiple modalities at the same time should theoretically be stronger than that of perturbing a single modality because it perturbs the inputs more making the predictions more distant from the true results. However, in practice, there may be results as shown in Fig. 5. For

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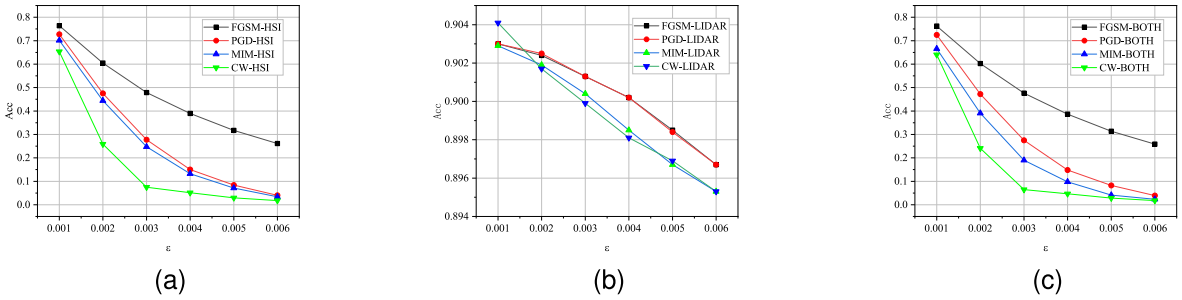


Fig. 3. Model accuracy under unimodal and multimodal attacks with different perturbation sizes. (a) Model accuracy under HSI unimodal attack with different perturbations. (b) Model accuracy under LiDAR unimodal attack with different perturbations. (c) Model accuracy under HSI-LiDAR multimodal attack with different perturbations.

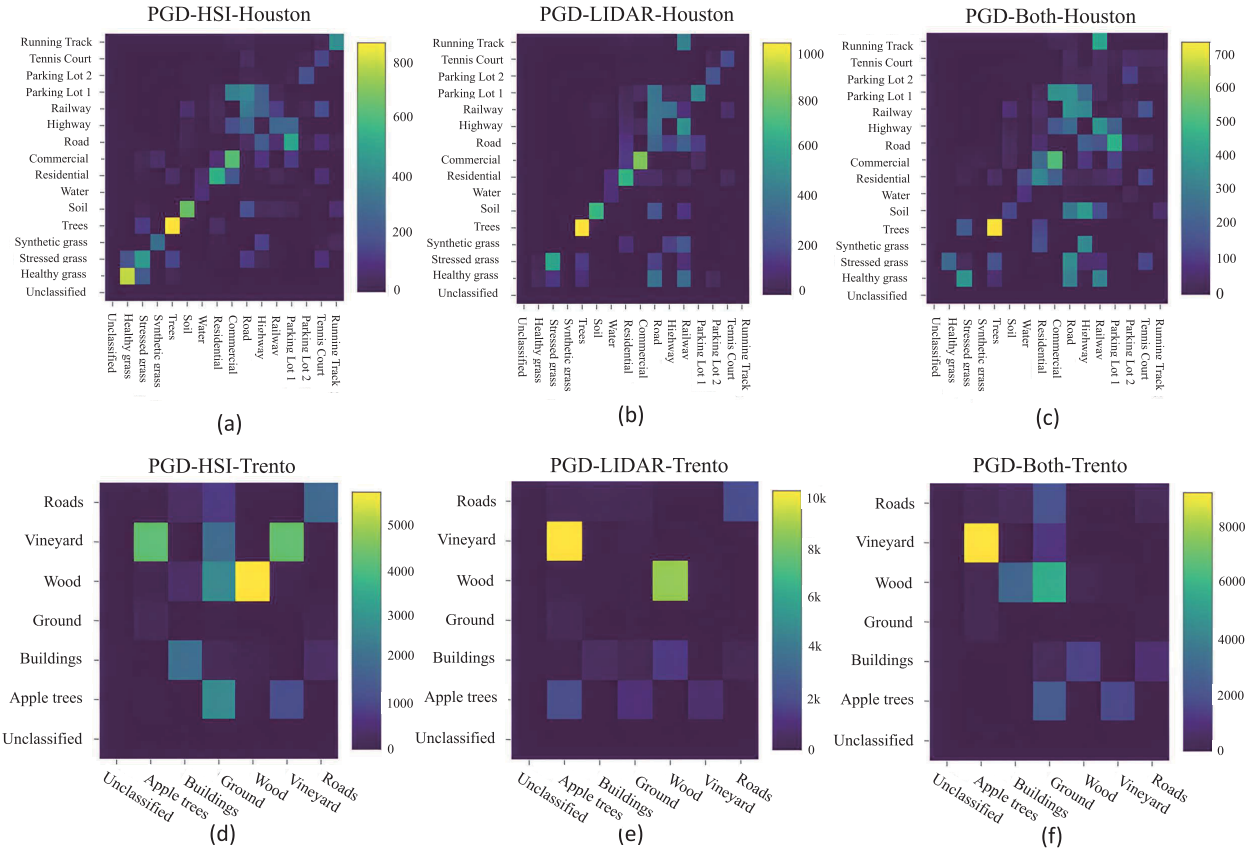


Fig. 4. Result maps under unimodal and multimodal attacks. (a)–(c) Result maps of unimodal attacks on HSI modality or LIDAR modality and multimodal attacks on both modalities by PGD attack under Houston dataset. (d)–(f) Result maps of unimodal attacks on HSI modality or LIDAR modality and multimodal attacks on both modalities by PGD attack under Trento dataset.

a multimodal model fusing HSI and LiDAR modalities, if only the HSI modality is perturbed, the model predicts successful misclassification of Vineyards from the Wood category, whereas if perturbation to the LiDAR modality is then added to the model on top of that, the model predicts back to the true category and the multimodal attack fails. We analyze the reason for this because the perturbations to both modalities are considered independently and there is no interaction between them, thus the two modal perturbations do not perturb the prediction in the same direction when they, respectively, move the prediction away from the true labels, which leads to conflicting effects of the attack, thus weakening the attack. To solve such a problem, we propose the cooperative adversarial loss, which tries to constrain the two modalities to perturb the results in the same direction.

2) Imbalance in the Size of Perturbations Added to Different Modalities: During a multimodal attack, one modality dominates in attacking the target model, which can lead to a situation where a large perturbation is added in HSI modality and relatively small perturbation in LiDAR modality. As shown in Fig. 5, in the multimodal attack against the multimodal model, the perturbation size added to the HSI modality after optimization of the attack algorithm is 0.0028, and the perturbation size added to the LiDAR modality is 0.0015. And since adversarial attack methods often have a limit on the maximum perturbation strength, this imbalance of perturbations between modalities is not conducive to an optimal attack. Therefore, we propose balanced perturbation loss to try to balance the size of perturbations between modalities.

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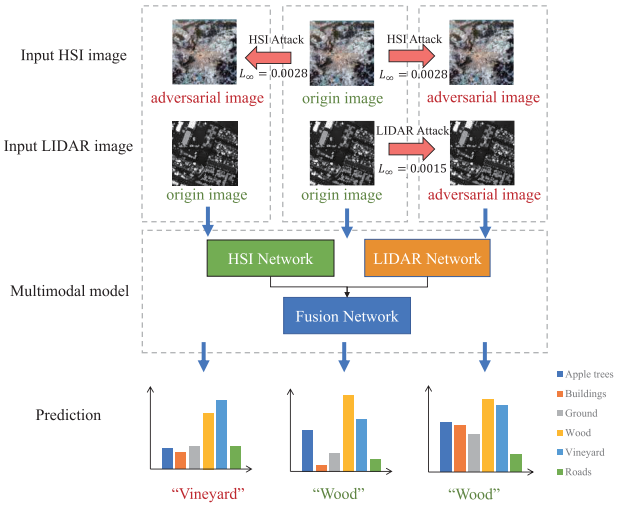


Fig. 5. Example of multimodal adversarial attack in multimodal remote sensing image classification.

B. Methods

The remote sensing multimodal model used in this study has two types of inputs, namely HSI and LiDAR data. The adversarial attack method used for this multimodal model is optimization-based and its optimized loss function is designed as follows. Here, L_p is the perturbation loss, the purpose of this design is to constrain the strength of the perturbation. The equation is as follows:

$$L_p = \left\|x'_H - x_H\right\|_2^2 + \left\|x'_L - x_L\right\|_2^2. \tag{18}$$

The loss L_f is an adversarial loss that maximizes the gap between the multimodal classification network's prediction and the actual label and encourages the prediction error of the network, where l_f is the cross-entropy loss

$$L_f = -l_f\left(f\left(x'_H, x'_L\right), y\right). \tag{19}$$

The loss L_{bp} is the balanced perturbation loss, which is designed to balance the size of the perturbation generated by each modality. $G_H(x_H)$ and $G_L(x_L)$ is the perturbation generated by each modality. T is the transfer function that aims to map the $G_H(x_H)$ and $G_L(x_L)$ vectors to the same dimension. We use the cosine similarity function l to assess the similarity of the perturbations of distinct modalities. To discover the best combination of perturbations, we hope that this loss can constrain the perturbations between different modalities to have similar distributions

$$L_{\text{bp}} = -l(G_H(x_H), T(G_L(x_L))). \tag{20}$$

The loss L_{ca} is the cooperative adversarial loss, which aims to make the perturbations of two modalities have similar perturbation directions on the results of the multimodal classification network, to avoid the conflicting effects of the two perturbations. This loss can help the method to find the optimal perturbation combination with smaller perturbation and more obvious adversarial perturbation effects, where l is the cosine similarity metric

$$L_{\text{ca}} = -l\left(f\left(x'_H, x_L\right) - f(x_H, x_L), f\left(x_H, x'_L\right) - f(x_H, x_L)\right). \tag{21}$$

TABLE VI

ACCURACY OF EARLY FUSION MULTIMODAL REMOTE SENSING CLASSIFICATION MODELS ON THE HOUSTON AND THE TRENTO DATASETS USING DIFFERENT ATTACK METHODS. THE ATTACK METHODS INCLUDE IMPROVED MULTIMODAL FGSM, MULTIMODAL PGD, MULTIMODAL MIM, MULTIMODAL C&W ATTACKS, AND THE MULTIMODAL MULTILOSS COOPERATIVE ATTACK PROPOSED IN THIS ARTICLE

Dataset	Attack	OA	AA	κ
Houston	Multi-FGSM	0.4760	0.5225	0.4308
	Multi-PGD	0.2750	0.2887	0.2115
	Multi-MIM	0.2438	0.2571	0.1776
	Multi-C&W	0.1839	0.2013	0.1138
	ours	0.1134	0.1238	0.0379
Trento	Multi-FGSM	0.6264	0.5751	0.5135
	Multi-PGD	0.4531	0.4083	0.3133
	Multi-MIM	0.4284	0.3842	0.2841
	Multi-C&W	0.2294	0.2195	0.0378
	ours	0.2198	0.2095	0.0278

TABLE VII

ACCURACY OF MIDDLE FUSION MULTIMODAL REMOTE SENSING CLASSIFICATION MODELS ON THE HOUSTON AND TRENTO DATASETS USING DIFFERENT ATTACK METHODS

Dataset	Attack	OA	AA	κ
Houston	Multi-FGSM	0.5475	0.5944	0.5086
	Multi-PGD	0.3412	0.3672	0.2842
	Multi-MIM	0.3056	0.3295	0.2457
	Multi-C&W	0.2037	0.2131	0.1362
	ours	0.1658	0.1730	0.0949
Trento	Multi-FGSM	0.7358	0.6139	0.6485
	Multi-PGD	0.5505	0.4650	0.4365
	Multi-MIM	0.5367	0.4581	0.4221
	Multi-C&W	0.3537	0.3482	0.2228
	ours	0.3420	0.3434	0.2104

The full loss for generating the adversarial perturbations is shown in the following equation, and the optimization objective is to minimize this loss function:

$$\min L_p + \alpha L_f + \beta L_{\text{bp}} + \gamma L_{\text{ca}} \tag{22}$$

where α, β, and γ are the weighting coefficients of each loss term, respectively. The architecture of the method is shown in Fig. 6.

C. Experiments

1) Experimental Settings: In this section, we use the same experimental settings as in Section III-B, with the same dataset and metrics, multimodal network architectures, and parameters.

In the following experiments, we test different metrics of the legitimate samples and adversarial samples on different multimodal fusion architectures using different attack methods. The hyperparameter α in (22) is set to 0.01 in the Houston dataset and 0.1 in the Trento dataset, and the hyperparameters β and γ are uniformly set to 0.01 and 1.

2) Comparison Experiments: Tables VI–VIII show the results of multimodal remote sensing image classification models subjected to different adversarial attack methods. We can observe that Multi-FGSM has the weakest attack capability, followed by Multi-MIM and Multi-PGD, Multi-C&W is slightly inferior to the proposed method in this article, and our proposed multimodal multiloss cooperative attack

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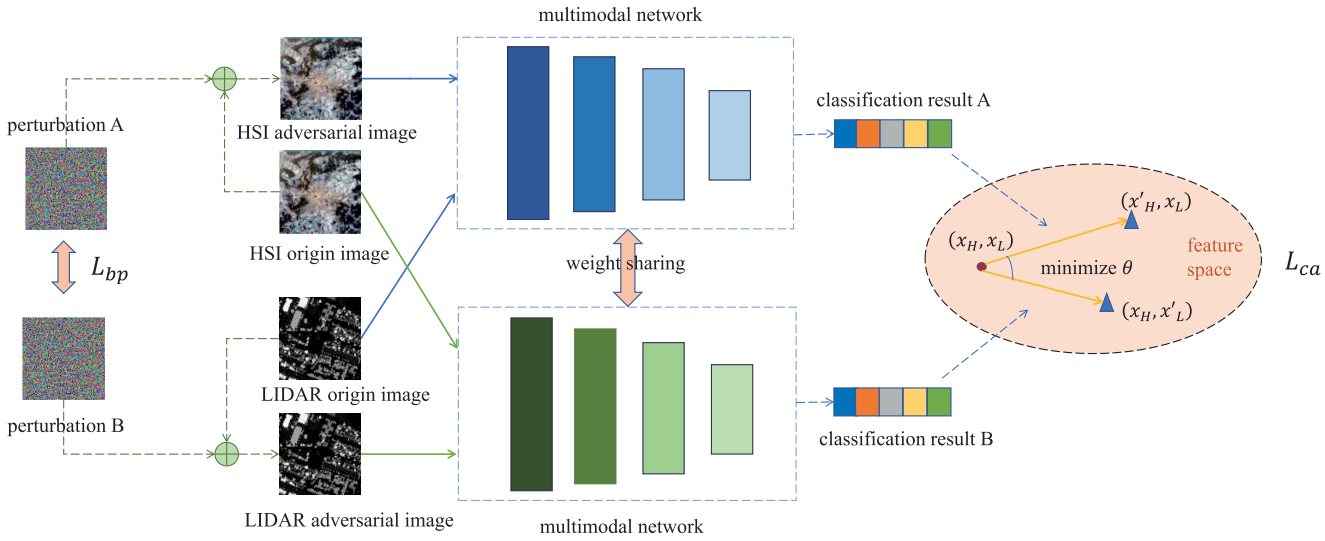


Fig. 6. Architecture of multimodal multiloss cooperative attack method.

TABLE VIII
ACCURACY OF LATE FUSION MULTIMODAL REMOTE SENSING CLASSIFICATION MODELS ON THE HOUSTON AND TRENTO DATASETS USING DIFFERENT ATTACK METHODS

Dataset	Attack	OA	AA	κ
Houston	Multi-FGSM	0.5161	0.5731	0.4756
	Multi-PGD	0.2961	0.3211	0.2373
	Multi-MIM	0.2593	0.2843	0.1976
	Multi-C&W	0.1941	0.1999	0.1270
	ours	0.1565	0.1586	0.0868
Trento	Multi-FGSM	0.7764	0.7207	0.7060
	Multi-PGD	0.5929	0.5210	0.4852
	Multi-MIM	0.5603	0.4950	0.4493
	Multi-C&W	0.2858	0.2828	0.1807
	ours	0.2677	0.2661	0.1634

method has the strongest attack performance among all the other methods in the baseline, which indicates the effectiveness of the proposed method.

The fusion strategy is also a potential factor that affects the performance and robustness of multimodal models. We compare the different fusion strategies early, middle, and late fusion through the experiments in Tables VI–VIII and find that the middle fusion is the most robust against multimodal attacks, followed by late fusion, and the early fusion method is the least robust. Even the more powerful multimodal multiloss cooperative attack we proposed still follows this pattern. This attack method is more effective against early fusion models, followed by late fusion models. It is least effective against middle fusion models. We can understand from the results that fusion at different modality feature levels is more robust than fusion at the data level and result level.

3) Visualization Experiments: From Fig. 7, we can see that the different multimodal adversarial attack methods of the early fusion multimodal model on the Trento dataset can cause most of the remote sensing image pixels to be misclassified, and the attacked multimodal target model cannot effectively resist the adversarial samples, among which the multimodal C&W and the multimodal multiloss cooperative attack methods proposed in this article have the highest success rate, which can reduce the accuracy of the model to about 20%. In general, the multimodal mutliloss cooperative attack method proposed in this article has a better attack success rate than other baseline methods.

4) Hyperparametric Analysis: Fig. 8 shows the effect of different β and γ parameter settings on the attack success rate of the multimodal multiloss cooperative attack method proposed in this article.

We set the values of β parameters in the interval of 0–0.1 and use a step size of 0.01 and γ parameters in the interval of 0 to 2 and use a step size of 0.2 to detect the effects of different parameter value settings. Fig. 8(a) shows the experimental data attacked by the multimodal multiloss cooperative adversarial attack method with different values of β on the models trained with early fusion, middle fusion, and late fusion strategies. We find that with the value of γ parameter set to 1, the models with β parameter between 0.01 and 0.06 have the lowest prediction accuracy, that is, the methods have the highest attack success rate on the models with the three fusion strategies. When the value of the β parameter is too high, exceeding 0.09, it will weaken the attack success rate of the method. The reason may be because too strong a constraint limits the diversity of perturbations thus reducing the probability of finding the best perturbation. Fig. 8(b) shows the experimental data of the multimodal cooperative adversarial attack method with different values of γ on the models trained by the early, middle, and late fusion strategies. We find that with the value of β parameter taken as 0.02, as the value of γ parameter increases, the stronger the method attacks. The performance of the attack algorithm proposed in this article is highly sensitive to the value setting of the two hyperparameters, so it is very important to adjust the value of the two hyperparameters reasonably.

5) Ablation Experiments: The role of the balanced perturbation loss aims to make the distribution of the perturbations added to the two different modalities HSI and

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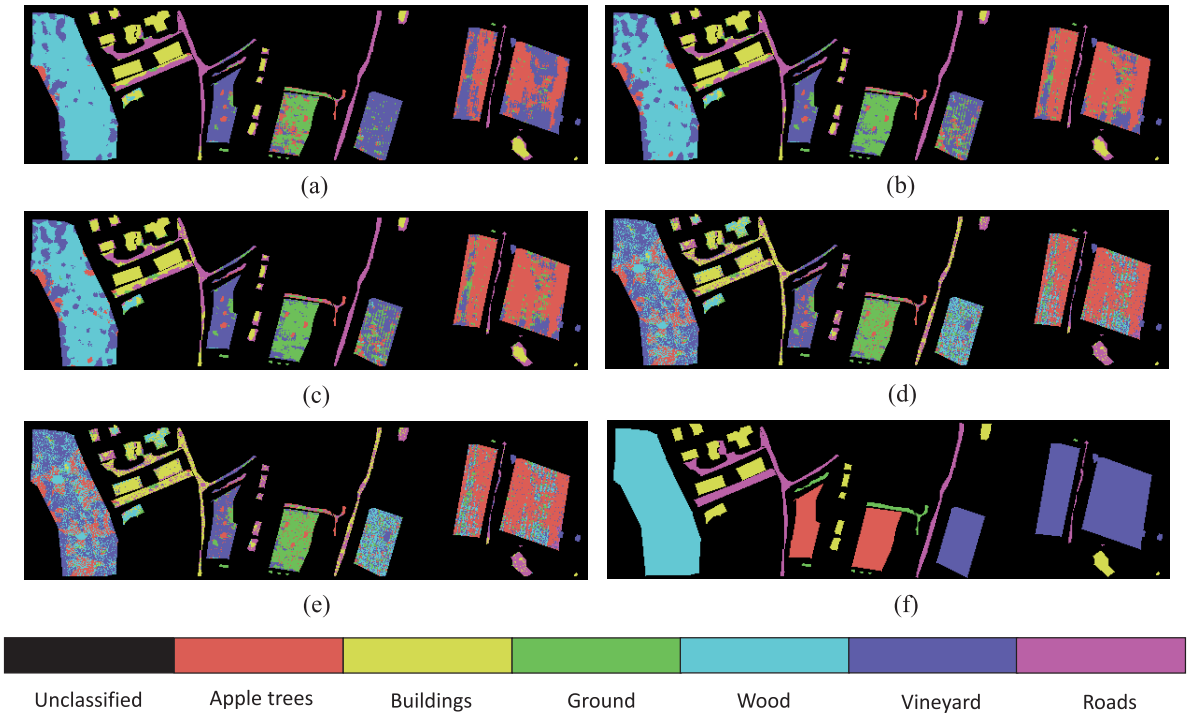


Fig. 7. Visualization of results of different multimodal attack methods on the early fusion model of the Trento dataset. (a)–(e) Classification images of adversarial samples under multimodal FGSM, multimodal PGD, multimodal MIM, multimodal C&W, and the multimodal multiloss cooperative attack proposed in this article, respectively, and (f) ground truth of the original image.

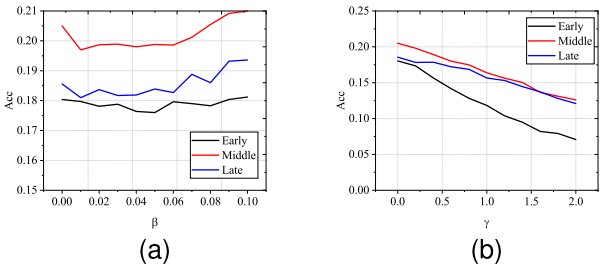


Fig. 8. Influence of β and γ parameters on the effectiveness of attacks. (a) Accuracy of different values of β. (b) Accuracy of different values of γ.

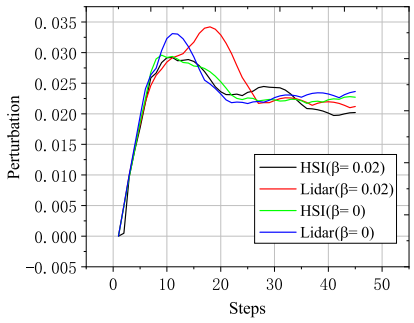


Fig. 9. Adding perturbation change curves in the optimization process. It shows the perturbation size curves added to the HSI and LiDAR of different modalities with the number of optimization iteration steps when the β parameter is 0 and 0.01.

LiDAR similar. Fig. 9 shows the perturbation size curves added to the HSI and LiDAR of different modalities with the number of optimization iteration steps when the β parameter is 0 and 0.02, that is, without or with the balanced perturbation loss term, respectively. We can see that the perturbation size tends to stabilize after 40 iteration steps. From the stabilized perturbations, we can see that the perturbation added to the HSI modality is slightly larger than that added to the LiDAR modality when the β parameter is either 0 or 0.02. The perturbation added to the HSI and LiDAR modalities when the β parameter is 0.02, that is, when the balanced perturbation loss term exists, is smaller than that added to the modality when the β parameter is 0, which suggests that the addition of balanced perturbation loss can help the method to optimize the perturbation size to a better effect.

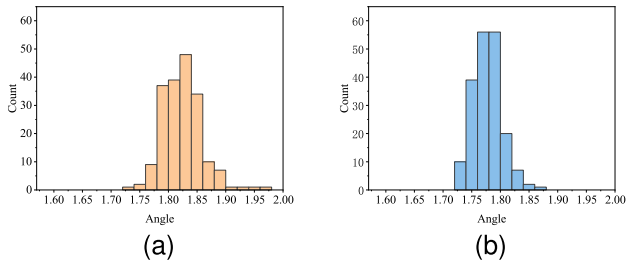


Fig. 10. Change vector angle statistics for prediction results against HSI modality attack and against LiDAR modality attack. (a) When the γ parameter is 0. (b) When the γ parameter is 2.

The role of the proposed cooperative adversarial perturbation loss is to synergize the effects of different modality perturbations on the classification results. From Fig. 10, we can see that the distribution of the angles of the perturbation results caused by the HSI modality attack and the LiDAR modality attack when the γ parameter is 0 is obviously more concentrated to the right than that when the

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TABLE IX

ABLATION EXPERIMENTS ON THE HOUSTON AND TRENTO DATASETS FOR DIFFERENT FUSION STRATEGIES. NORMAL DENOTES THE OPTIMIZATION METHOD CONTAINING ONLY L_p LOSS AND L_f LOSS

Dataset	Attack	Early	Middle	Late
Houston	Normal	0.1804	0.2050	0.1856
	Normal+L_{bp}	0.1781	0.1987	0.1837
	Normal+L_{ca}	0.0708	0.1260	0.1209
	Normal+ L_{bp} +L_{ca}	0.0699	0.1220	0.1200
Trento	Normal	0.2296	0.3413	0.2683
	Normal+L_{bp}	0.2226	0.3402	0.2664
	Normal+L_{ca}	0.2115	0.3365	0.2665
	Normal+L_{bp}+L_{ca}	0.2101	0.2075	0.2664

γ parameter is 2, that is, the angle of the perturbation results is generally larger, which indicates that the addition of the cooperative adversarial perturbation loss does reduce the angle of the vectors of the perturbation results of the two modalities.

From Table IX, we can see that the addition of the balanced perturbation loss L_{bp} and the cooperative adversarial loss L_{ca} enhances the attack strength of the multimodal cooperative attack method, which proves the reliability of the proposed two theories. We also find that the cooperative adversarial loss is more effective in optimizing the attack effect than the balanced perturbation loss.

V. CONCLUSION

In this article, we investigate the robustness of multimodal remote sensing image classification models and propose a new multimodal multiloss cooperative adversarial attack method. We test the robustness of different models against different adversarial attack methods and different numbers of attack modalities under different fusion strategies and show that multimodal models are also susceptible to adversarial attacks. Meanwhile, our proposed multimodal multiloss cooperative adversarial attack method adopts the balanced perturbation loss and cooperative adversarial loss to balance the perturbation size of different modalities and coordinate attacks of different modalities to prevent attack effect conflict. This method achieves better attack strength than other baseline methods. We hope that this study will draw academic attention to the robustness of multimodal models. In the future, we hope that more research can be conducted on designing methods to deal with multimodal attacks in various scenarios and improving the robustness of multimodal models to defend against these adversarial attacks.

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